Hands-on example

Deploy weather agent to Cloud Run

Step 1: Create agent

Shell
```
mkdir weather_agent_cloud_run
cd weather_agent_cloud_run
```

Create `agent.py`:

Python
```python
from google.adk.agents import LlmAgent

root_agent = LlmAgent(
    model='gemini-2.5-flash',
    name='weather_agent',
    description='Weather agent on Cloud Run',
    instruction='''
    You are a helpful weather assistant.
    Provide weather information for cities worldwide.
    '''
)
```

Create `__init__.py`:

Python
```python
from .agent import root_agent
```

Step 2: Deploy

Shell
```
export GOOGLE_CLOUD_PROJECT="your-project-id"
export GOOGLE_CLOUD_LOCATION="us-central1"

adk deploy cloud_run \
    --project=$GOOGLE_CLOUD_PROJECT \
    --region=$GOOGLE_CLOUD_LOCATION \
    --service_name=weather-agent \
    --with_ui \
    weather_agent_cloud_run/
```

Expected output:

```None
Building container...
Deploying to Cloud Run...
✓ Service deployed

Service URL: https://weather-agent-xyz123.us-central1.run.app
```

Step 3: Test

Open the service URL in your browser to access the web UI (similar to `adk web`).

Accessing deployed agents

Web UI (with `--with_ui`)

Navigate to service URL in browser:

```None
https://weather-agent-xyz123.us-central1.run.app
```

Interactive chat interface—share URL with team for testing.

API access

```Shell
SERVICE_URL="https://weather-agent-xyz123.us-central1.run.app"

curl -X POST $SERVICE_URL/api/query \
    -H "Content-Type: application/json" \
    -d '{"input": "What is the weather in Seattle?"}'
```

Python access

```python
import requests

SERVICE_URL = "https://weather-agent-xyz123.us-central1.run.app"

response = requests.post(
    f"{SERVICE_URL}/api/query",
    json={"input": "What's the weather in London?"}
)

print(response.json())
```

Session service note

Important: Cloud Run does NOT automatically configure session service like Agent Engine.

For production persistence, configure VertexAiSessionService in your agent:

```python
from google.adk.sessions import VertexAiSessionService

session_service = VertexAiSessionService(
    project=os.environ.get('GOOGLE_CLOUD_PROJECT'),
    location=os.environ.get('GOOGLE_CLOUD_LOCATION')
)
```

Comparison

Platform	Session Service
Agent Engine	Automatic
Cloud Run	Manual configuration

Key takeaways

You've now deployed an agent to Cloud Run, giving you an alternative deployment path with more flexibility. Cloud Run is particularly useful when you need a web UI deployed alongside your agent or have custom container requirements. For most standard ADK agents, Agent Engine remains the simpler choice.

Cloud Run basics

- `adk deploy cloud_run` packages and deploys agent
- `--with_ui` includes web interface
- Public HTTPS URL for access

When to use Cloud Run

- Want web UI deployed with agent
- Custom container requirements
- Existing Cloud Run infrastructure

When to use Agent Engine

- Standard ADK agents (simpler)
- Automatic session/memory services
- Prefer fully managed infrastructure

Same agent, different platforms

Your agent code works on both Agent Engine and Cloud Run—just different deployment commands.